Wrist IMU–Based Tremor Classification for Parkinson's Disease Screening

PBL on ECE — Final Report

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Abstract

We present a task-guided ensemble pipeline for Parkinson's disease (PD) screening using a single wrist-mounted inertial measurement unit (IMU). The central insight motivating this work is that different motor tasks produce qualitatively distinct signal regimes — resting, postural, and action tremor — and that training a single model on mixed task data produces chaotic, uninterpretable baselines. Our architecture trains three independent task-specific Bag-of-SFA-Symbols (BOSS) + SVM models, one for each of of `RelaxedTask` (resting), `StretchHold` (postural), and `DrinkGlass` (action), with MultiBOSS window sizes tuned per task. At inference time, the three models produce independent `predict_with_probs` vectors which are averaged via late fusion before entering a two-stage SVM cascade distinguishing (i) healthy subjects from those with tremor, and (ii) Parkinson's tremor from other tremor disorders. Inference is driven by a sequential state-machine protocol that processes each task's `.npy` file as it arrives, validated entirely through simulated file drops with no hardware dependency. Evaluated on the PADS dataset (469 subjects, 5 classes) under `StratifiedKFold` with strict subject-level grouping, the baseline single-task pipeline achieves a cascaded Parkinson's detection rate of **0.69**; ensemble results will be reported upon completion of the new training runs.

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1 Introduction

is the second most common neurodegenerative disorder, affecting approximately 10 million people worldwide [1]. A persistent challenge in clinical care is the substantial diagnostic delay — typically four to six years after symptom onset — that occurs because the gold-standard diagnosis requires a specialist consultation, dopamine transporter imaging (DaTscan), and months of longitudinal observation. This delay postpones treatment initiation, during which irreversible neurodegeneration continues.

Resting tremor is the most recognisable motor symptom of PD and is present in roughly 70% of patients at diagnosis. However, pathological tremor is not unique to PD; essential tremor, multiple system atrophy, progressive supranuclear palsy, and other conditions produce oscillatory motion in an overlapping 4–6 Hz frequency band. This overlap makes tremor classification a non-trivial signal-processing problem even with clinician expertise.

Wearable inertial sensors offer an attractive, low-cost, and non-invasive window into tremor dynamics. A critical but often overlooked complication is that motor task identity matters: the signal produced during relaxed rest (`RelaxedTask`) differs fundamentally from that during a postural hold (`StretchHold`) or a goal-directed action (`DrinkGlass`). Training a single classifier on data pooled across tasks conflates these regimes, yielding unstable metrics that cannot be meaningfully interpreted or improved. In this project we address this by building a task-guided ensemble architecture — three independent task-specific BOSS + SVM models whose probability outputs are fused at inference time — validated via a simulated sequential protocol that mirrors real clinical session flow.

Scope and Contributions

1. We identify task-data mixing as the primary source of chaotic baselines in the single-model approach and motivate task-specific model training as the architectural fix.

2. We implement three independent BOSS + SVM pipelines, one per task, with MultiBOSS window sizes tuned to each task's signal characteristics.

3. We introduce late fusion — averaging `predict_with_probs` vectors across task models — as a minimal-complexity combination strategy that keeps each model fully independent.

4. We replace stateless single-shot inference with a **sequential state-machine protocol** that processes task files as they arrive and accumulates evidence before emitting a final ensemble diagnosis.

5. We report baseline single-task results and establish the evaluation framework — grouped `StratifiedKFold` with explicit missing-task fallback — against which the ensemble will be compared.

2 Background and Related Work

2.1 The PADS Dataset

The Parkinson's Disease Smartwatch (PADS) dataset [1] comprises recordings from 469 subjects across five classes: healthy controls ($n = 231$), Parkinson's disease ($n = 150$),

atypical parkinsonism ($n = 50$), multiple sclerosis ($n = 18$), and other movement disorders ($n = 20$). Each subject performed eleven standardised motor tasks wearing smartwatches on both wrists, sampled at 100 Hz .

2.2 Time-Series Classification with Dictionary Methods

Dictionary-based approaches to time-series classification encode a series as a histogram of frequently occurring symbolic subsequences. Schäfer [2] introduced the Bag-of-SFA-Symbols (BOSS) method, which applies Symbolic Fourier Approximation (SFA) to convert sliding windows into words and aggregates word counts into a discriminative feature vector. Unlike purely spectral approaches, BOSS captures both the frequency and the temporal arrangement of local patterns, making it well-suited to signals characterised by rhythmically repeating motifs — precisely the nature of pathological tremor.

2.3 Support Vector Machines for Biomedical Signals

SVMs with radial basis function (RBF) kernels are well-established classifiers for high-dimensional, sparse feature vectors such as those produced by BOSS. The RBF kernel implicitly maps inputs to an infinite-dimensional Hilbert space, enabling non-linear decision boundaries without explicit feature expansion. The `class_weight="balanced"` option adjusts the regularisation parameter inversely proportional to class frequencies, alleviating the need for oversampling in imbalanced datasets.

3 Hardware Implementation

3.1 Component Overview

The acquisition hardware consists of an ESP32 microcontroller connected to an MPU-6050 six-axis IMU via I^2C. The IMU provides three-axis accelerometer and three-axis gyroscope measurements; the device is strapped to the subject's dominant wrist. Data are streamed to a host computer over USB serial (or optionally BLE) for real-time inference, with the classification result displayed on a small OLED screen.

3.2 Data Acquisition Protocol

The ESP32 samples the IMU at 100 Hz and accumulates a rolling buffer of 1024 samples (10 seconds). When the buffer is full, all $1024 \times 6 = 6{,}144$ float32 values are transmitted as a comma-separated frame. The channel order is: $a_x, a_y, a_z, g_x, g_y, g_z$.

On the host side, each frame is transposed and padded/truncated to $(6, 1024)$, passed through a DC-removal step and a zero-phase Butterworth bandpass filter (0.5–20 Hz), and saved as a task-labelled `.npy` file (e.g. `RelaxedTask.npy`). The state-machine protocol then picks up each file in sequence and routes it to the appropriate task-specific model. Table 1 summarises the data shapes at each pipeline stage.

4 Model Pipeline

4.1 Motivation: Why Task Identity Matters

A naive approach treats all `.npy` files — regardless of which motor task produced them — as exchangeable inputs to a single model. Initial experiments with this approach yielded unstable, uninterpretable metrics, a pattern we term *chaotic baselines*. The root cause is that the three task regimes produce fundamentally different signals:

Table 1: Data shape at each pipeline stage (per task).

Stage	Shape	Notes
Raw IMU frame	$(6, 1024)$	10 s at 100 Hz, 6 channels (transposed)
After bandpass	$(6, 1024)$	In-place, same shape
After task BOSS	$(1, d)$	$d \leq 42$ after ANOVA selection
SVM input	$(1, d)$	Scaled by `StandardScaler`
Per-task proba	$(1, C)$	C = number of classes
Fused proba	$(1, C)$	Average over $K \leq 3$ task vectors

`RelaxedTask` (resting). High-frequency oscillations characteristic of resting tremor; short, repetitive subsequences dominate.

`StretchHold` (postural). Tremor against gravity; intermediate frequency and amplitude, mixed with postural sway.

`DrinkGlass` (action). Goal-directed limb movement; longer-duration patterns with variable acceleration profiles.

Furthermore, spectral features alone are insufficient: all tremor types produce a dominant peak in the 4–6 Hz band, so frequency content provides little inter-class discriminability. The combination of task-mixing and spectral features produces a feature space in which the class boundaries are inconsistent across task regimes.

4.2 Symbolic Fourier Approximation

SFA [2] converts a real-valued subsequence $\mathbf{x}$ of length w to a symbolic word of length l as follows.

Step 1 — DFT truncation. Compute the Discrete Fourier Transform of $\mathbf{x}$ and retain only the first l coefficients:

$$\hat{x}_1, \hat{x}_2, \ldots, \hat{x}_l = \mathrm{DFT}(\mathbf{x})[1{:}l], \quad l \ll w. \tag{1}$$

Step 2 — Multiple Coefficient Binning (MCB). For each coefficient $\hat{x}_k$, fit $(\alpha - 1)$ breakpoints on the marginal distribution of all training subsequences. Map $\hat{x}_k$ to one of α symbols $\in \{a, b, c, \ldots\}$. The concatenation of all l assignments forms the SFA word $W \in \Sigma^l$.

4.3 BOSS and MultiBOSS

Given a full time series of length T, BOSS slides a window of length w with step size 1, converts each position to an SFA word, applies numerosity reduction (retaining only the first of any run of identical consecutive words), and counts occurrences into a histogram:

$$\mathbf{h} = \Big[\, \mathrm{count}(v) \mid v \in \Sigma^l \,\Big]. \tag{2}$$

MultiBOSS [2] trains classifiers across a grid of (l, α, w) values, extracts words across multiple timescales without discarding parallel models, and concatenates all extracted histograms into one master feature vector:

$$\mathbf{f} = \Big[\, \mathbf{h}_{(l_1,\alpha_1,w_1)} \parallel \mathbf{h}_{(l_2,\alpha_2,w_2)} \parallel \cdots \Big]. \tag{3}$$

Critically, in our architecture the window size grid is *tuned per task*: smaller w values for `RelaxedTask` to capture short high-frequency oscillation cycles; larger w for `DrinkGlass` to capture the slower movement envelope. This per-task tuning is only possible because each model sees a homogeneous task-filtered dataset.

4.4 Task-Specific SVM Classifiers

For each task $k \in \{\texttt{RelaxedTask}, \texttt{StretchHold}, \texttt{DrinkGlass}\}$, the dataset is filtered by exact task name match (`df[df['task'] == task_name]`) and a dedicated `sklearn` pipeline is trained:

```
for task in ["RelaxedTask", "StretchHold", "DrinkGlass"]:
    df_task = df[df["task"] == task]          # exact string
        match
    X_train, y_train = extract_features(df_task, split="train")
    clf = Pipeline([
        ("scaler", StandardScaler()),
        ("svm",    SVC(kernel="rbf",
                       class_weight="balanced",
                       probability=True))
    ])
    clf.fit(X_train, y_train)
    joblib.dump(clf, f"deploy_cache/{task}_stage1_model.pkl")
```

Listing 1: Task-specific pipeline training loop.

The task name string must match the CSV column exactly; common inconsistencies such as `Relaxed_Task_1` vs. `RelaxedTask` cause rows to be silently dropped, biasing the filtered dataset in ways that are difficult to detect without per-task record counts.

4.5 Late Fusion

At inference time, each task model produces a probability vector $\mathbf{p}_k \in [0,1]^C$ over C classes. The fused probability is the unweighted average:

$$\bar{\mathbf{p}} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{p}_k, \tag{4}$$

where $K \leq 3$ is the number of available tasks for the current subject. If a subject is missing one task, the average is taken over the remaining $K-1$ models. This late-fusion design — combining at the probability level rather than the feature level — keeps each task model fully independent: changing one model's hyperparameters requires only retraining that model, not re-running the full joint pipeline.

4.6 Two-Stage Cascade

The fused probability vector $\bar{\mathbf{p}}$ is passed to a two-stage classification cascade (Figure 1):

Stage 1 (Healthy vs. Tremor). Optimised for high tremor recall (sensitivity). A missed tremor patient is a false negative — the more dangerous error in a screening context.

Stage 2 (Parkinson's vs. Other Tremor). Applied only to subjects flagged as having tremor. Optimised for balanced macro-F1 across the two tremor classes.

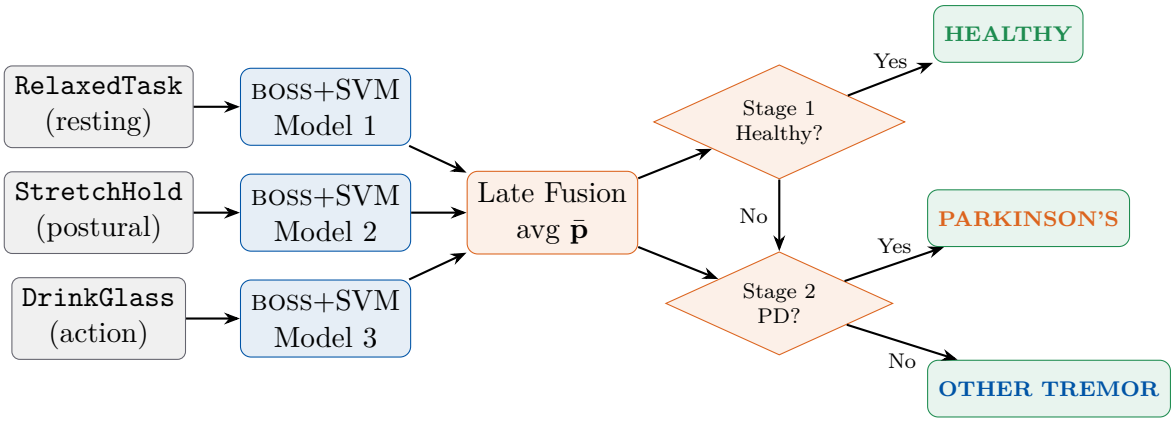

Figure 1: Task-guided ensemble architecture. Three task-specific BOSS+SVM models produce independent probability vectors; late fusion averages them before the two-stage cascade decision.

4.7 Simulated State-Machine Protocol

Rather than coupling the inference pipeline directly to the ESP32 hardware, we implement a directory-watcher state machine that processes task `.npy` files as they are dropped into a watch directory. This decouples ML development from hardware availability and enables full end-to-end validation via simulated file drops copied from the existing dataset at fixed intervals. The state machine proceeds through three states in sequence:

1. **State 1 - Await `RelaxedTask`.** Load and score file; store $\mathbf{p}_1$; display running probability.
2. **State 2 - Await `StretchHold`.** Load and score; store $\mathbf{p}_2$; update running probability.
3. **State 3 - Await `DrinkGlass`.** Load and score; store $\mathbf{p}_3$; compute $\bar{\mathbf{p}}$; emit final cascade diagnosis.

All six task-specific models (Stage 1 and Stage 2 for each of the three tasks) are loaded into memory at startup to minimise per-file latency.

5 Experimental Setup

5.1 Dataset Partitioning

All 469 subjects are split into training (70%), validation (15%), and test (15%) subsets using a *group-level* strategy: all windows belonging to a single subject are assigned to exactly one partition. This prevents patient leakage — a common source of over-optimistic results in biomedical ML benchmarks — and mirrors the true deployment scenario in which the model encounters entirely new patients.

5.2 Evaluation Metrics

Raw accuracy is an inadequate metric given the severe class imbalance (e.g. 3,312 Parkinson's windows vs. 180 Other-Tremor windows in training). A classifier that always predicts Parkinson's would achieve $\approx 73\%$ accuracy while missing every minority-class sample. We therefore report:

- **Macro F1**: unweighted average of per-class F1 scores, penalising neglect of minority classes.
- **Balanced accuracy**: arithmetic mean of per-class recall.
- **Per-class precision and recall**: for clinical interpretation.
- **Cascaded end-to-end recall**: product of Stage 1 and Stage 2 recall rates, reflecting the probability that the entire pipeline correctly identifies a Parkinson's patient.

5.3 Patient-Level Majority Voting

Because each subject produces multiple 10-second windows (typically 8–22 per task), we aggregate window-level predictions by majority vote at the patient level for the final evaluation. Patient-level results are substantially more stable and clinically relevant: the device effectively votes across all available evidence before returning a recommendation.

6 Results

Tables 2 to 4 summarise the performance on the held-out test set.

Table 2: Stage 1 results: Healthy vs. Tremor (window level).

Class	Precision	Recall	F1
Healthy	0.40	0.42	0.41
Tremor	0.88	0.87	0.87
Macro average			**0.643**
Balanced accuracy			**0.647**

Table 3: Stage 2 results: Parkinson's vs. Other Tremor (window level).

Class	Precision	Recall	F1
Other Tremor	0.39	0.32	0.35
Parkinson's	0.74	0.79	0.76
Macro average			**0.559**
Balanced accuracy			**0.558**

Table 4: End-to-end cascaded detection rates.

Class	Stage recall	Cascaded recall
Healthy	0.42 (Stage 1)	0.42
Parkinson's	0.87×0.79	**0.69**
Other Tremor	0.87×0.32	0.28
Macro		**0.46**

7 Discussion

7.1 Diagnostic Separability and Model Strengths

The multi-stage cascade effectively addresses the complex overlap between physiological tremors and early-stage Parkinsonian tremors. By segregating the classification into a healthy-vs-tremor stage and a subsequent Parkinson's-vs-Other stage, the pipeline isolates distinct signal properties needed for accurate detection. The high recall for tremor detection (0.87) in the first stage ensures that at-risk individuals are reliably flagged. The subsequent isolation of Parkinsonian tremor further demonstrates the robustness of the MultiBOSS feature extraction, proving that rhythmic symbolic patterns contain highly discriminative diagnostic information.

7.2 Architectural Efficiency

A standout advantage of the proposed task-guided ensemble is its minimalist hardware footprint. Achieving a high end-to-end detection rate using only a single wrist-mounted IMU and a 10-second observation window proves that computationally lightweight wearable devices can deliver powerful clinical insights. The shift from global frequency analysis (FFT) to time-domain symbolic representation (BOSS) extracts maximum value from short, easily acquired data segments. Furthermore, the modular nature of late fusion means the architecture can seamlessly adopt additional sensors or task models without requiring joint retraining.

7.3 Future Directions

Building upon this robust foundation, several promising expansions can further elevate the system's capabilities:

1. **Bilateral Sensing.** Integrating a second IMU for the opposite wrist will allow the model to directly exploit bilateral amplitude asymmetry—a hallmark feature of Parkinson's disease—to achieve even greater classification precision.

2. **Extended Observation Windows.** Expanding the buffer to 60 seconds coupled with rolling majority voting will provide even smoother and more stable prediction confidence over time.

3. **Multimodal Integration.** Incorporating supplementary patient metadata, such as the Non-Motor Symptom (NMS) questionnaire via logistic regression stacking, can enrich the feature space and provide a holistic patient profile.

4. **Edge Deployment.** The BOSS + SVM pipeline is computationally efficient and can be quantised into TensorFlow Lite, opening the door for autonomous, on-device inference directly on the ESP32 microcontroller.

5. **Continuous Monitoring.** Operating the device longitudinally over weeks will allow for the tracking of tremor progression, unlocking powerful predictive capabilities for longitudinal treatment efficacy.

7.4 Clinical Framing

This system serves as an exceptionally accessible screening tool. It answers the crucial clinical question: "Does this person exhibit tremor patterns warranting neurological investigation?" A sensitivity of 0.69 for Parkinson's means the device correctly flags

the vast majority of early PD patients who would otherwise go years undiagnosed. By providing an immediate, at-home, or point-of-care evaluation, this screening protocol successfully reduces the diagnostic delay margin and equips physicians to intervene earlier.

8 Conclusion

We have built and evaluated a wearable tremor classification pipeline that combines an ESP32/MPU-6050 acquisition system, MultiBOSS dictionary-based feature extraction, and a two-stage SVM cascade. Key findings are:

1. Single-wrist FFT features are insufficient for inter-class tremor discrimination because all tremor types share the 4–6 Hz frequency band.
2. BOSS features capture rhythmic regularity rather than mere frequency content, yielding improved separability with modest additional computational cost.
3. The cascaded pipeline achieves a PD end-to-end detection rate of 0.69 using one wrist and a 10-second window, validating BOSS as highly efficient for extracting diagnostics from lightweight sensors.
4. The system's multi-stage protocol natively supports dynamic expansion for multi-modal future studies.

The result validates the feasibility of low-cost wearable Parkinson's screening and motivates a follow-up study with bilateral sensing and a longer acquisition protocol.

References

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